

4.2 Irrational Numbers



LESSON FOCUS

Identify and order irrational numbers.

The room below the rotunda in the Manitoba Legislative Building is the Pool of the Black Star. It has a circular floor.

Make Connections

The formulas for the area and circumference of a circle involve π , which is not a rational number because it cannot be written as a quotient of integers.

What other numbers are not rational?

Construct Understanding

TRY THIS

Work with a partner.

These are rational numbers.	These are not rational numbers.
$\sqrt{100}$ $\sqrt{0.25}$ $\sqrt[3]{8}$ 0.5	$\sqrt{0.24}$ $\sqrt[3]{9}$ $\sqrt{2}$
$\frac{5}{6}$ $\sqrt{\frac{9}{64}}$ 0.8^2 $\sqrt[5]{-32}$	$\sqrt{\frac{1}{3}}$ $\sqrt[4]{12}$

- A.** How are radicals that are rational numbers different from radicals that are not rational numbers?

- B.** Which of these radicals are rational numbers?
Which are not rational numbers? How do you know?

$$\sqrt{1.44}, \sqrt{\frac{64}{81}}, \sqrt[3]{-27}, \sqrt{\frac{4}{5}}, \sqrt{5}$$

- C.** Write 3 other radicals that are rational numbers.
Why are they rational?
- D.** Write 3 other radicals that are not rational numbers.
Why are they not rational?

Radicals that are square roots of perfect squares, cube roots of perfect cubes, and so on are rational numbers. Rational numbers have decimal representations that either terminate or repeat.

Irrational Numbers

An **irrational number** *cannot* be written in the form $\frac{m}{n}$, where m and n are integers, $n \neq 0$. The decimal representation of an irrational number neither terminates nor repeats.

When an irrational number is written as a radical, the radical is the *exact value* of the irrational number; for example, $\sqrt{2}$ and $\sqrt[3]{-50}$. We can use the square root and cube root keys on a calculator to determine *approximate values* of these irrational numbers.

$\sqrt{2}$

1.414213562

$\sqrt[3]{-50}$

-3.684031499

There are other irrational numbers besides radicals; for example, π .

We can approximate the location of an irrational number on a number line.

If we do not have a calculator, we use perfect powers to estimate the value.

For example, to locate $\sqrt[3]{-50}$ on a number line,

we know that $\sqrt[3]{-27} = -3$ and $\sqrt[3]{-64} = -4$.

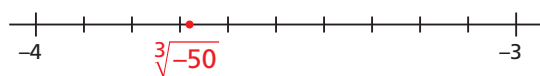
Guess: $\sqrt[3]{-50} \doteq -3.6$

Test: $(-3.6)^3 = -46.656$

Guess: $\sqrt[3]{-50} \doteq -3.7$

Test: $(-3.7)^3 = -50.653$

This is close enough to represent on a number line.



Since $(-3.7)^3 = -50.653$, then $\sqrt[3]{-50}$ is slightly greater than -3.7 , so mark a point to the right of -3.7 on the number line.

Example 1 Classifying Numbers

Tell whether each number is rational or irrational.
Explain how you know.

a) $-\frac{3}{5}$ b) $\sqrt{14}$ c) $\sqrt[3]{\frac{8}{27}}$

SOLUTION

a) $-\frac{3}{5}$ is rational since it is written as a quotient of integers.

Its decimal form is -0.6 , which terminates.

b) $\sqrt{14}$ is irrational since 14 is not a perfect square.

The decimal form of $\sqrt{14}$ neither repeats nor terminates.

c) $\sqrt[3]{\frac{8}{27}}$ is rational since $\frac{8}{27}$ is a perfect cube.

$\sqrt[3]{\frac{8}{27}} = \frac{2}{3}$ or $0.\overline{6}$, which is a repeating decimal

CHECK YOUR UNDERSTANDING

1. Tell whether each number is rational or irrational. Explain how you know.

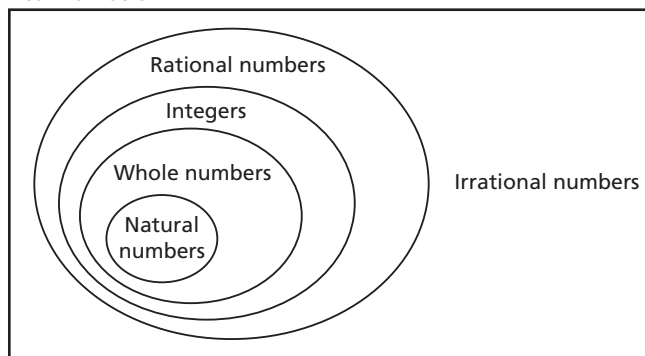
a) $\sqrt{\frac{49}{16}}$ b) $\sqrt[3]{-30}$
c) 1.21

[Answers: a) rational
b) irrational c) rational]

Together, the rational numbers and irrational numbers form the set of **real numbers**.

This diagram shows how these number systems are related.

Real Numbers



Example 2 Ordering Irrational Numbers on a Number Line

Use a number line to order these numbers from least to greatest.

$\sqrt[3]{13}$, $\sqrt{18}$, $\sqrt{9}$, $\sqrt[4]{27}$, $\sqrt[3]{-5}$

(Solution continues.)

SOLUTION

13 is between the perfect cubes 8 and 27, and is closer to 8.

$$\begin{array}{ccc} \sqrt[3]{8} & \sqrt[3]{13} & \sqrt[3]{27} \\ \downarrow & \downarrow & \downarrow \\ 2 & ? & 3 \end{array}$$

Use a calculator.

$$\sqrt[3]{13} = 2.3513...$$

$3\sqrt[3]{13}$

2.351334688

18 is between the perfect squares 16 and 25, and is closer to 16.

$$\begin{array}{ccc} \sqrt{16} & \sqrt{18} & \sqrt{25} \\ \downarrow & \downarrow & \downarrow \\ 4 & ? & 5 \end{array}$$

Use a calculator.

$$\sqrt{18} = 4.2426...$$

$\sqrt{18}$

4.242640687

$$\sqrt{9} = 3$$

27 is between the perfect fourth powers 16 and 81, and is closer to 16.

$$\begin{array}{ccc} \sqrt[4]{16} & \sqrt[4]{27} & \sqrt[4]{81} \\ \downarrow & \downarrow & \downarrow \\ 2 & ? & 3 \end{array}$$

Use a calculator.

$$\sqrt[4]{27} = 2.2795...$$

$4\sqrt[4]{27}$

2.279507057

-5 is between the perfect cubes -1 and -8, and is closer to -8.

$$\begin{array}{ccc} \sqrt[3]{-1} & \sqrt[3]{-5} & \sqrt[3]{-8} \\ \downarrow & \downarrow & \downarrow \\ -1 & ? & -2 \end{array}$$

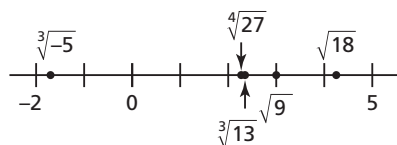
Use a calculator.

$$\sqrt[3]{-5} = -1.7099...$$

$3\sqrt[3]{-5}$

-1.709975947

Mark each number on a number line.



From least to greatest: $\sqrt[3]{-5}$, $\sqrt[4]{27}$, $\sqrt[3]{13}$, $\sqrt{9}$, $\sqrt{18}$

CHECK YOUR UNDERSTANDING

2. Use a number line to order these numbers from least to greatest.

$$\sqrt{2}, \sqrt[3]{-2}, \sqrt[3]{6}, \sqrt{11}, \sqrt[4]{30}$$

[Answer: $\sqrt[3]{-2}$, $\sqrt{2}$, $\sqrt[3]{6}$, $\sqrt[4]{30}$, $\sqrt{11}$]

How can you order a set of irrational numbers if you do not have a calculator?

Discuss the Ideas

1. How do you determine whether a radical represents a rational or an irrational number? Use examples to explain.
2. How can you determine whether the decimal form of a radical represents its exact value?

Exercises

A

3. Tell whether each number is rational or irrational.

a) $\sqrt{12}$	b) $\sqrt[4]{16}$
c) $\sqrt[3]{-100}$	d) $\sqrt{\frac{4}{9}}$
e) $\sqrt{1.25}$	f) 1.25
4. Classify each number below as:

a) a natural number	b) an integer
c) a rational number	d) an irrational number

 $\frac{4}{3}, 0.\overline{34}, -5, \sqrt[4]{9}, -2.1538, \sqrt[3]{27}, 7$

B

5. a) Why are $\sqrt{49}$ and $\sqrt[4]{16}$ rational numbers?
b) Why are $\sqrt{21}$ and $\sqrt[3]{36}$ irrational numbers?
6. Look at this calculator screen.

$\sqrt{\{150\}}$
12.24744871

- a) Is the number 12.247 448 71 rational or irrational? Explain.
- b) Is the number $\sqrt{150}$ rational or irrational? Explain.
7. a) Sketch a diagram to represent the set of rational numbers and the set of irrational numbers.
b) Write each number that follows in the correct set.
 $\frac{1}{2}, -\sqrt{3}, \sqrt{4}, \sqrt[4]{5}, -\frac{7}{6}, \sqrt[3]{8}, 10.12, -13.\overline{4}, \sqrt{0.15}, \sqrt{0.16}, 17$
8. For which numbers will the cube root be irrational? Use 2 different strategies to justify your answers.

a) 8	b) 64	c) 30	d) 300
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9. Sketch a number line for each irrational number and label its approximate location. Explain your reasoning.
a) $\sqrt{5}$ b) $\sqrt[3]{12}$ c) $\sqrt[4]{25}$ d) $\sqrt[3]{-12}$
10. Use a number line to order the irrational numbers in each set from greatest to least.
a) $\sqrt[3]{70}, \sqrt{50}, \sqrt[4]{100}, \sqrt[3]{400}$
b) $\sqrt{89}, \sqrt[4]{250}, \sqrt[3]{-150}, \sqrt[3]{150}$
11. Use a number line to order these numbers from least to greatest. How can you verify your answer?
 $\sqrt{40}, \sqrt[3]{500}, \sqrt{98}, \sqrt[3]{98}, \sqrt{75}, \sqrt[3]{300}$
12. Use a number line to order these numbers from least to greatest. Identify which numbers are irrational and which are rational.
 $\frac{-14}{5}, \frac{123}{99}, -2, \sqrt[3]{-10}, \sqrt{4}$
13. How do you use irrational numbers when you calculate the length of the hypotenuse of a right triangle with legs 5 cm and 3 cm?
14. a) Which of the following statements are true? Explain your reasoning.

i) All natural numbers are integers.
ii) All integers are rational numbers.
iii) All whole numbers are natural numbers.
iv) All irrational numbers are roots.
v) Some rational numbers are natural numbers.

 b) For each statement in part a that is false, provide examples to explain why.
15. Write a number that is:

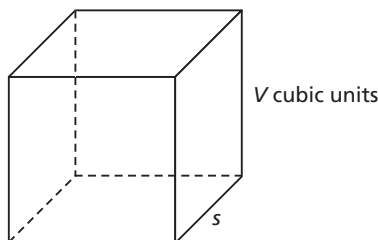
a) a rational number but not an integer
b) a whole number but not a natural number
c) an irrational number

16. a) Create a diagram to show how these number systems are related: irrational numbers; rational numbers; integers; whole numbers; and natural numbers. Write each number below in your diagram.

$\frac{3}{5}$, 4.91919, 16, $\sqrt[3]{-64}$, $\sqrt{60}$, $\sqrt{9}$, -7 , 0

- b) Choose 3 more numbers that would be appropriate for each number system in your diagram. Include these numbers.

17. This diagram shows a cube with volume V cubic units and edge length s units.



Provide a value of V for which s is:

- a) irrational b) rational

18. The *golden rectangle* appears in art and architecture. It has the property that the ratio of its length to width is $\frac{1 + \sqrt{5}}{2}$ to 1. The shape of the front of the Parthenon, in Greece, is a golden rectangle.



- a) Use a calculator. Write the value of $\frac{1 + \sqrt{5}}{2}$ to the nearest tenth.
 b) Use the number from part a as the length in inches of a rectangle. Draw a golden rectangle.
 c) Measure other rectangles in your classroom. Do any of these rectangles approximate a golden rectangle? Justify your answer.

19. The ratio $\frac{1 + \sqrt{5}}{2}$: 1 is called the *golden ratio*.

Use the dimensions of the Great Pyramid of Giza, from Chapter 1, page 26. Show that the ratio of its base side length to its height approximates the golden ratio.

20. Determine whether the perimeter of each square is a rational number or an irrational number. Justify your answer.

- a) a square with area 40 cm^2
 b) a square with area 81 m^2

C

21. Suppose you know that $\sqrt[n]{\frac{a}{b}}$ is a rational number and that a and b have no common factors. What can you say about the prime factorizations of a and b ?
22. For each description, sketch and label a right triangle or explain why it is not possible to create a triangle.
- a) All sides have rational number lengths.
 b) Exactly 2 sides have rational number lengths.
 c) Exactly 1 side has a rational number length.
 d) No sides have rational number lengths.
23. a) Can the square root of a rational number be irrational? Show your reasoning.
 b) Can the square root of an irrational number be rational? Show your reasoning.
24. Describe strategies to generate numbers with the property that their square roots, cube roots, and fourth roots are all rational numbers. Support your strategies with examples.

Reflect

Describe strategies you can use to determine if a radical represents a rational number or an irrational number.